The Logic of Consumer Choice
Chapter 20

Utility Theory

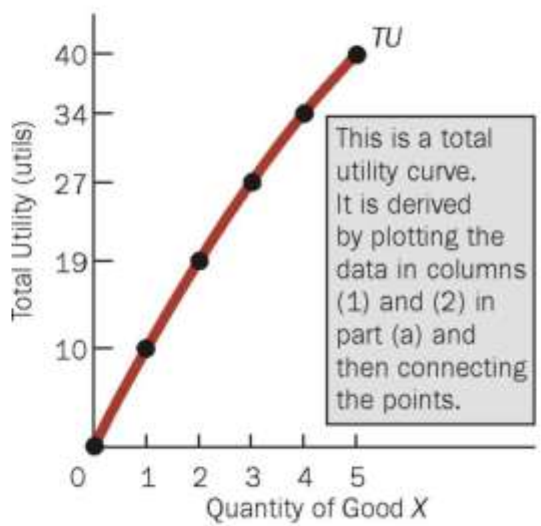
- A good that gives you ***Utility*** is one that has the power to satisfy wants, or that gives you satisfaction.
- Utils are an artificial construct used to “measure” utility.
- **Total Utility** is the total satisfaction a person receives from consuming a particular quantity of good.
- **Marginal Utility** is the additional utility gained from consuming an additional unit of some good.
- **The law of Diminishing Marginal Utility** states that for a given time period, the marginal utility gained by consuming equal successive units of a good will decline as the amount consumed increases

Total Utility, Marginal Utility, and the Law of Diminishing Utility

The law of diminishing marginal utility is based on the idea that if a good has a variety of uses but only one unit of the good is available, then the consumer will use the first unit to satisfy his or her most urgent want.

(1) Units of Good X	(2) Total Utility (utils)	(3) Marginal Utility (utils)
0	0	—
1	10	10
2	19	9
3	27	8
4	34	7
5	40	6

(a)



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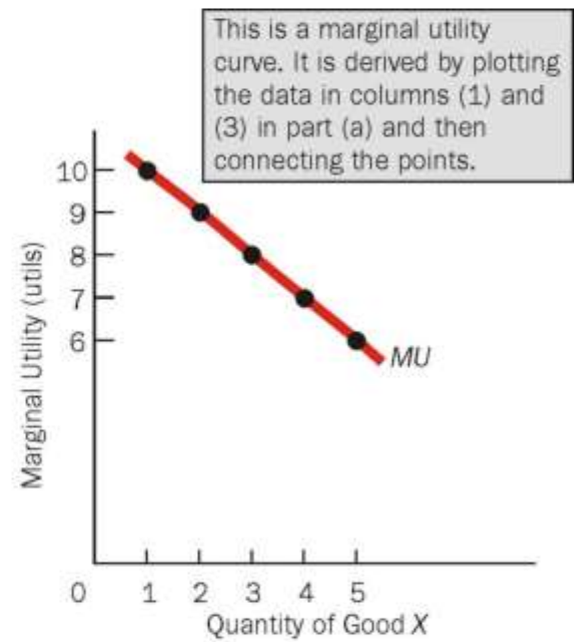


(c)

Interpersonal Utility Comparison

- The utility obtained by one person cannot be scientifically or objectively compared with the utility obtained from the same thing by another person because utility is subjective.

Consumer equilibrium

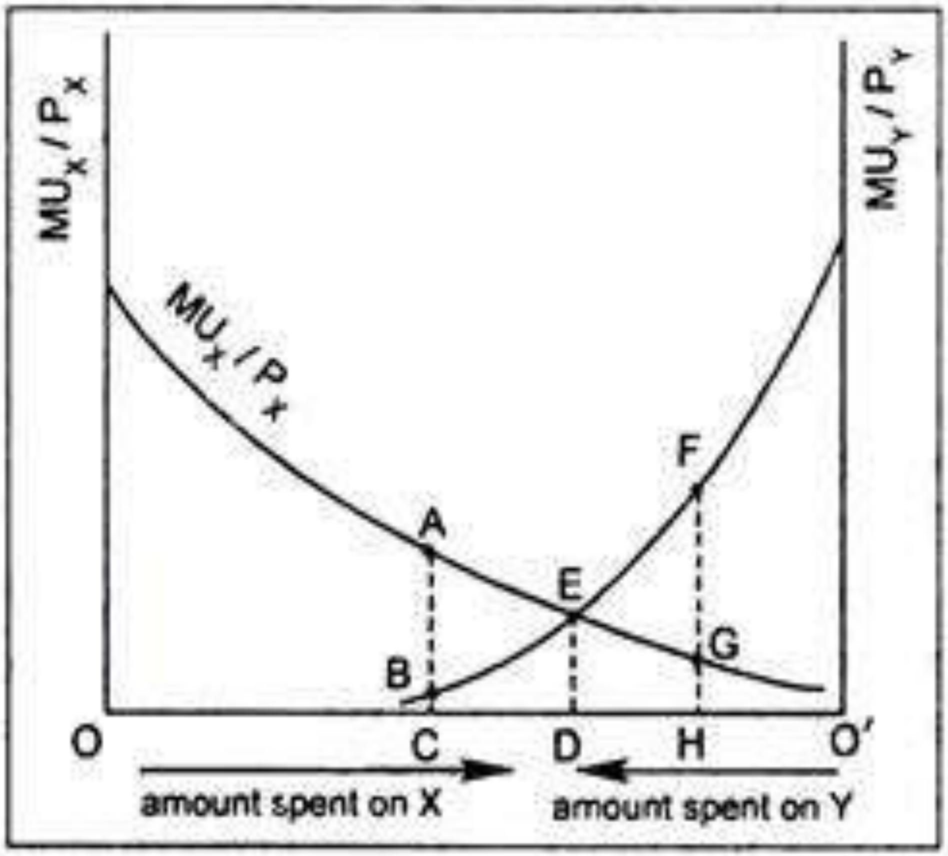
- Marginal utility begins to decline with the second unit of a good consumed.
- The marginal utility associated with consume equal successive units of a good will eventually decline as the amount consumed increases.
- A consumer is in equilibrium when he or she derives the same marginal utility per dollar for all goods. This is known as the Equi-marginal principle.

The equilibrium occurs when the consumer has spent all of his or her income and the marginal utilities per dollar spent on each good purchased are equal:

$$MU_A/P_A = MU_B/P_B \ldots\ldots\ldots\ldots\ldots = MU_z/P_z$$

where the letters A–Z represent all the goods a person buys

Equi-marginal principle



Equi-marginal Principle

Maximizing utility and the law of Demand

Suppose a consumer of oranges and apples is currently in equilibrium; that is,

$MU_A/P_A = Mu_0/P_0$

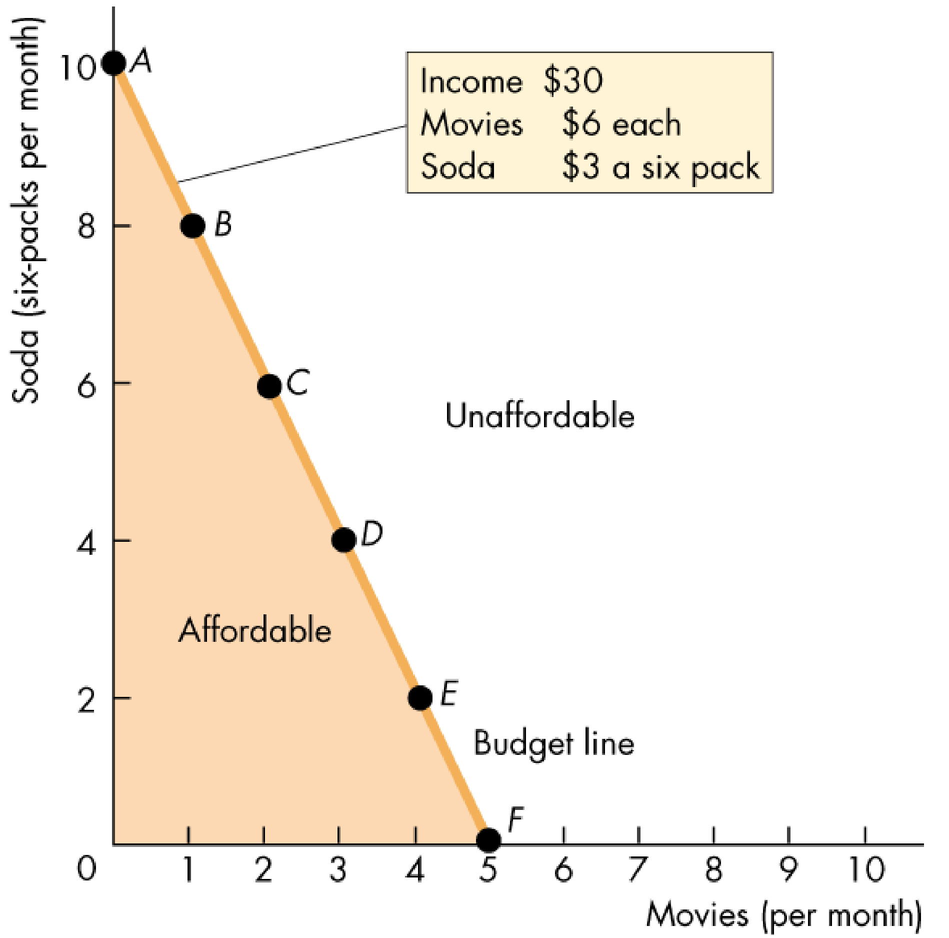
When in equilibrium, the consumer is maximizing utility. Now, suppose the price of oranges falls.

Then the situation becomes $MU_0/P_0 > MU_A/P_A$

The consumer will attempt to restore equilibrium by buying more oranges. This behavior—buying more oranges when their price falls—is consistent with the law of demand.

Consumption Possibilities

– Figure shows a consumer’s budget line.



Divisible goods can be bought in any quantity desired along the budget line (gasoline, for example)

Indivisible goods must be bought in whole units at the points marked on the budged line (movies, for example).

Consumption Possibilities

- The Budget Equation
  - We can describe the budget line by using a budget equation
  - The budget equation states that
    - Expenditure = Income
  - Call the price of soda P_S, the quantity of soda Q_S, the price of a movie P_M, the quantity of movies Q_M, and income Y.
  - Lisa's budget equation is:
    - $P_S Q_S + P_M Q_M = Y.$

Consumption Possibilities

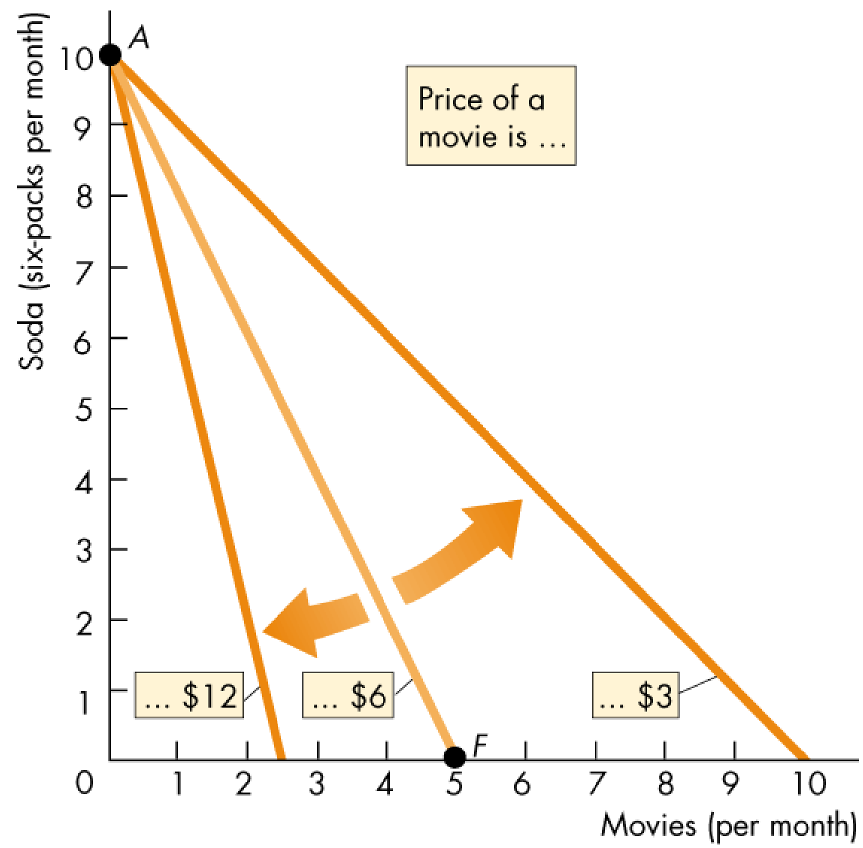
- $P_x Q_x + P_y Q_y = M$
- Divide both sides of this equation by P_S, to give:
  - $Q_y + (P_x/P_y)Q_x = M/P_y$
- Then subtract $(P_M/P_S)Q_M$ from both sides of the equation to give:
  - $Q_y = M/P_y - (P_x/P_y)Q_x$
- The term M/P_y is Lisa's real income in terms of Y.
- The term P_x/P_y is the relative price of a X in terms of Y.

Consumption Possibilities

- A household’s real income is the income expressed as a quantity of goods the household can afford to buy.
- Lisa’s real income in terms of soda is the point on her budget line where it meets the *y*-axis.
- A relative price is the price of one good divided by the price of another good.
- It is the magnitude of the slope of the budget line
- The relative price shows how many sodas must be forgone to see an additional movie.

Consumption Possibilities

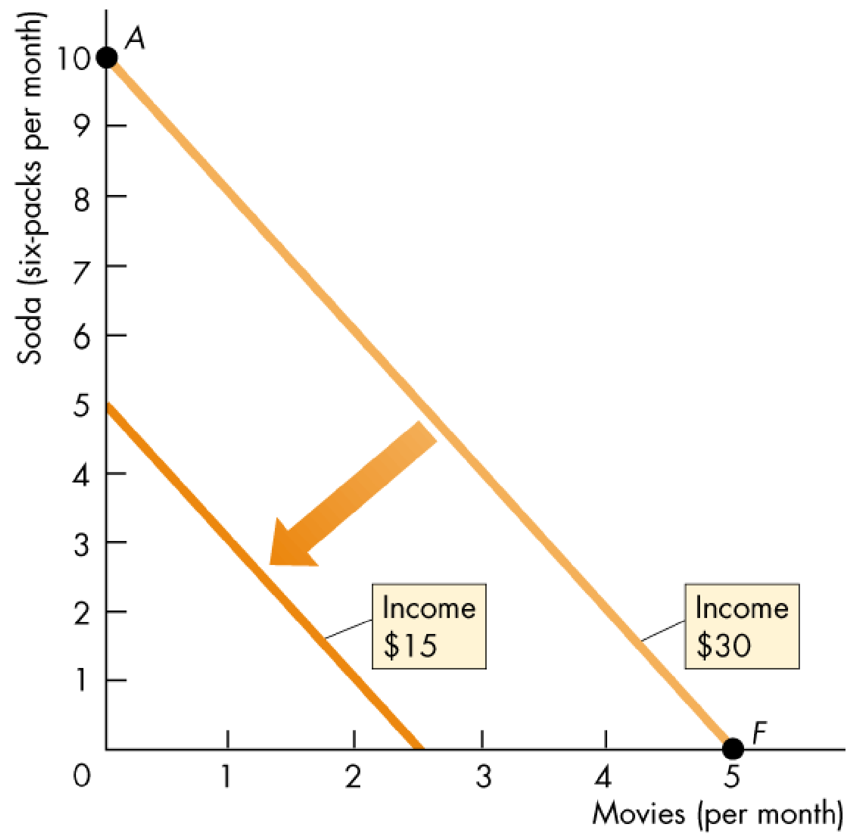
- A fall in the price of the good on the *x*-axis increases the affordable quantity of that good and decreases the *slope* of the budget line.
- Figure shows the rotation of a budget line after a change in the relative price of movies.



(a) A change in price

Consumption Possibilities

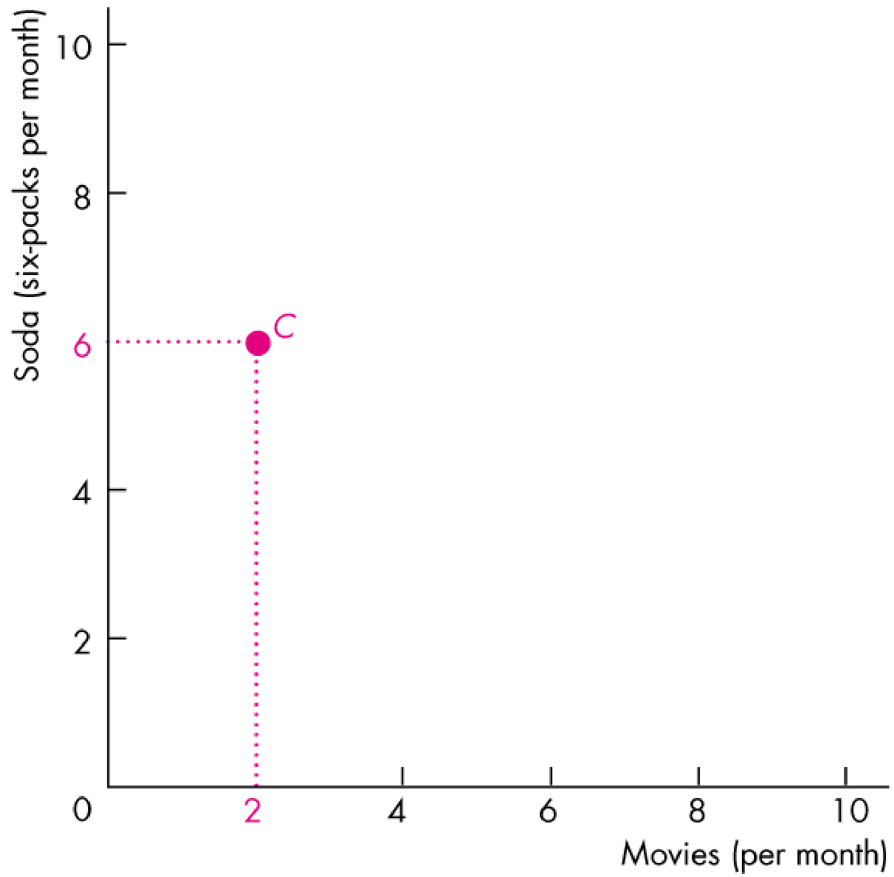
- An change in the household's income brings a parallel shift of the budget line.
- The slope of the budget line doesn't change because the relative price doesn't change.
- Figure shows the effect of a fall in income.



(b) A change in income

Preferences and Indifference Curves

- An **indifference curve** is a line that shows combinations of goods among which a consumer is *indifferent*.
- Figure illustrates a consumer's indifference curve.
- At point *C*, Lisa consumes 2 movies and 6 six-packs a month.

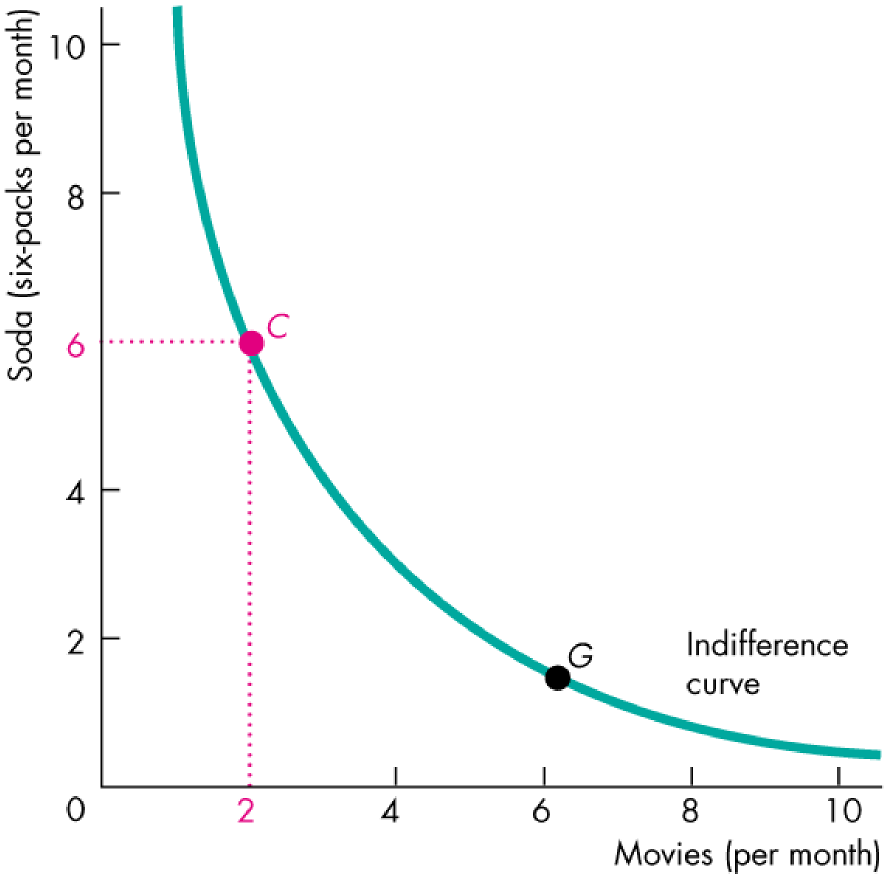


(a) An indifference curve

Preferences and Indifference Curves

- Lisa can sort all possible combinations of goods into three groups: preferred, not preferred, and indifferent.
- An indifference curve joins all those points that Lisa says are just as good as *C*.

G is such a point. Lisa is indifferent between *C* and *G*.

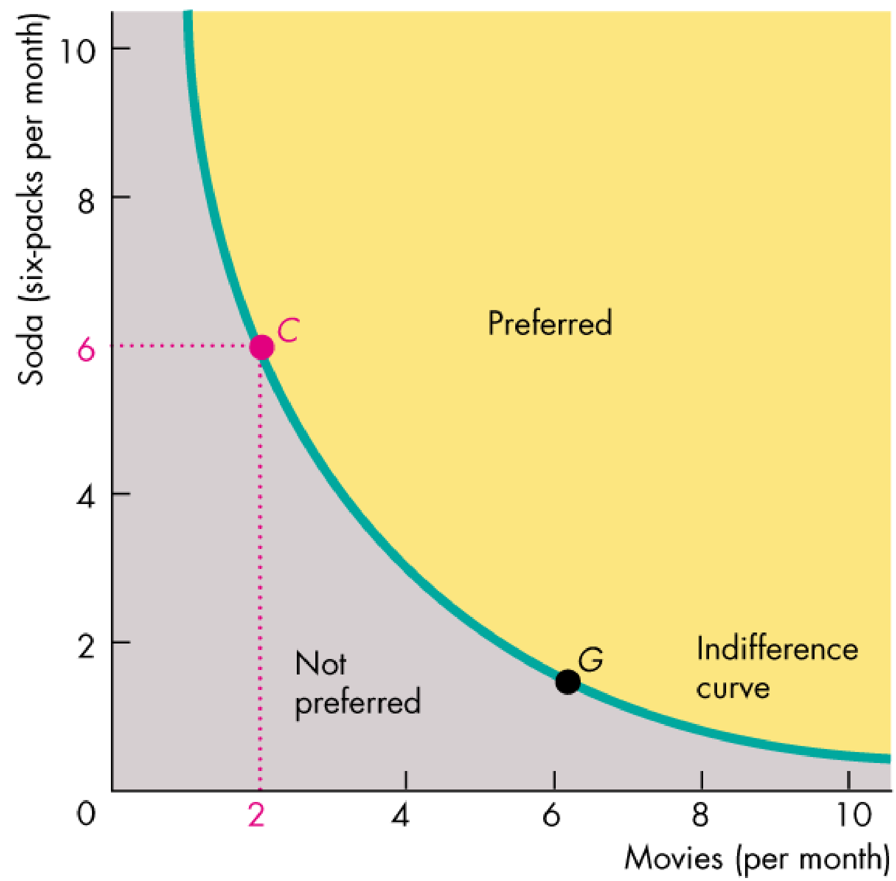


(a) An indifference curve

Preferences and Indifference Curves

– All the points above the indifference curve are preferred to the points on the curve.

And all the points on the indifference curve are preferred to the points below the curve.



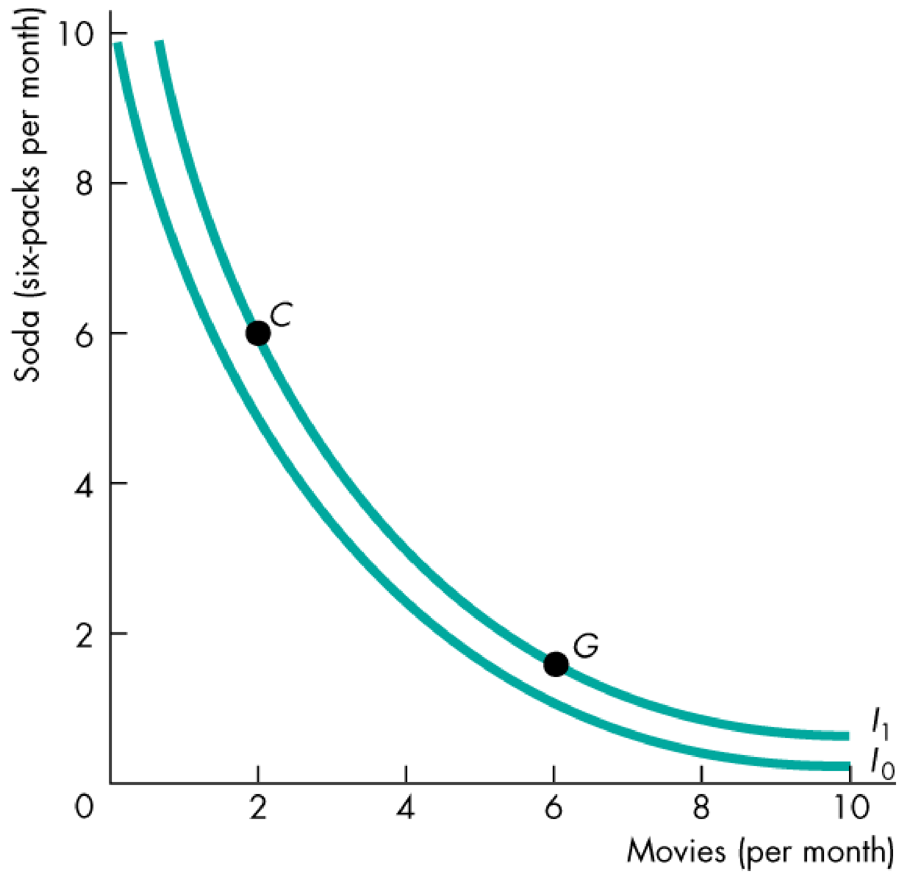
(a) An indifference curve

Preferences and Indifference Curves

- The indifference curve in the figure is just one of many curves that form an indifference map.

Call the indifference curve that we've just seen I_1.

An indifference curve below I_1 is I_0. All the points on I_1 are preferred to those on I_0.

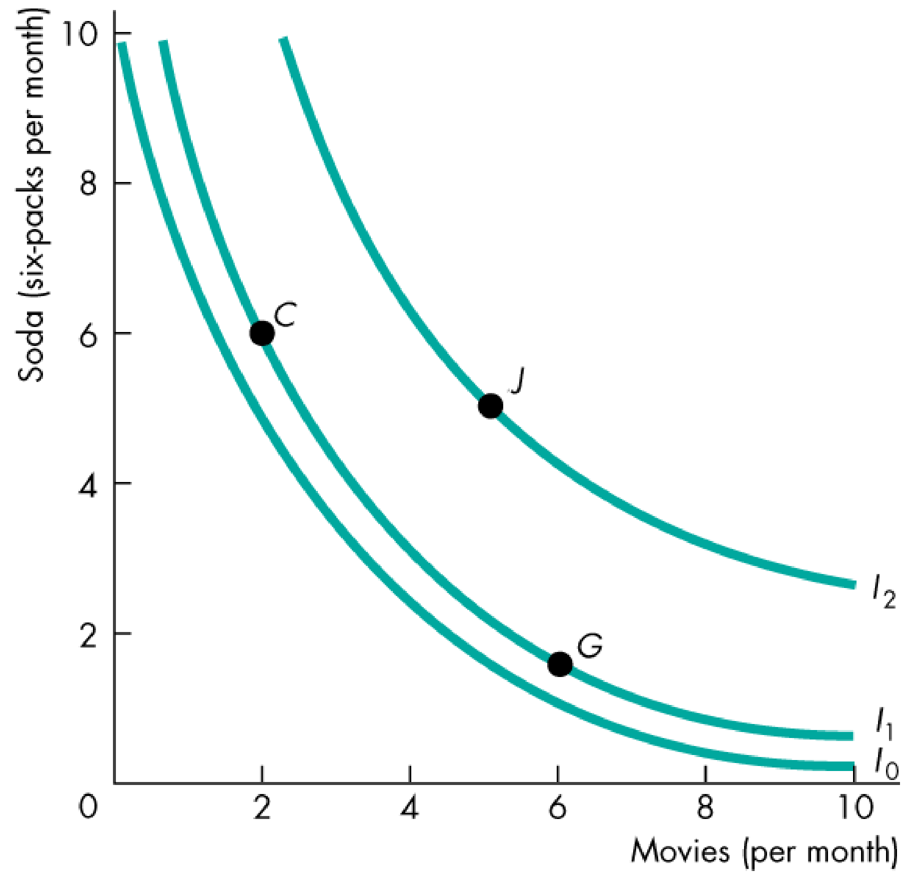


(b) Lisa's preference map

Preferences and Indifference Curves

- An indifference curve above I_1 is I_2. All the points on I_2 are preferred to those on I_1.

For example, point *J* is preferred to either point *C* or point *G*.

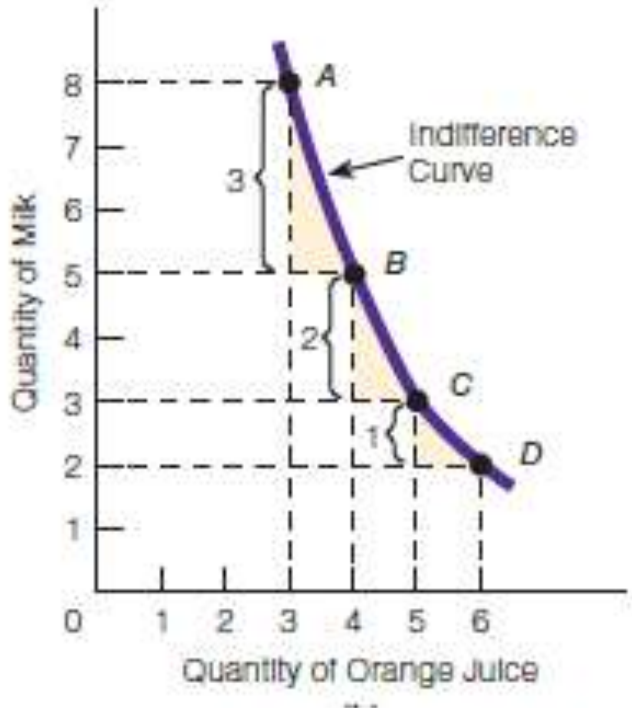


(b) Lisa's preference map

Indifference Curves & MRS

- Marginal Rate of Substitution
  - The **marginal rate of substitution**, (*MRS*) measures the rate at which a person is willing to give up good *y*, (the good measured on the *y*-axis) to get an additional unit of good *x* (the good measured on the *x*-axis) and at the same time remain indifferent (remain on the same indifference curve).
  - The magnitude of the slope of the indifference curve measures the marginal rate of substitution.

Marginal rate of substitution



Absolute value of the slope of the indifference curve = Marginal rate of substitution

$$= \frac{MU_{\text{good on horizontal axis}}}{MU_{\text{good on vertical axis}}}$$

Law of diminishing Marginal Rate of Substitution

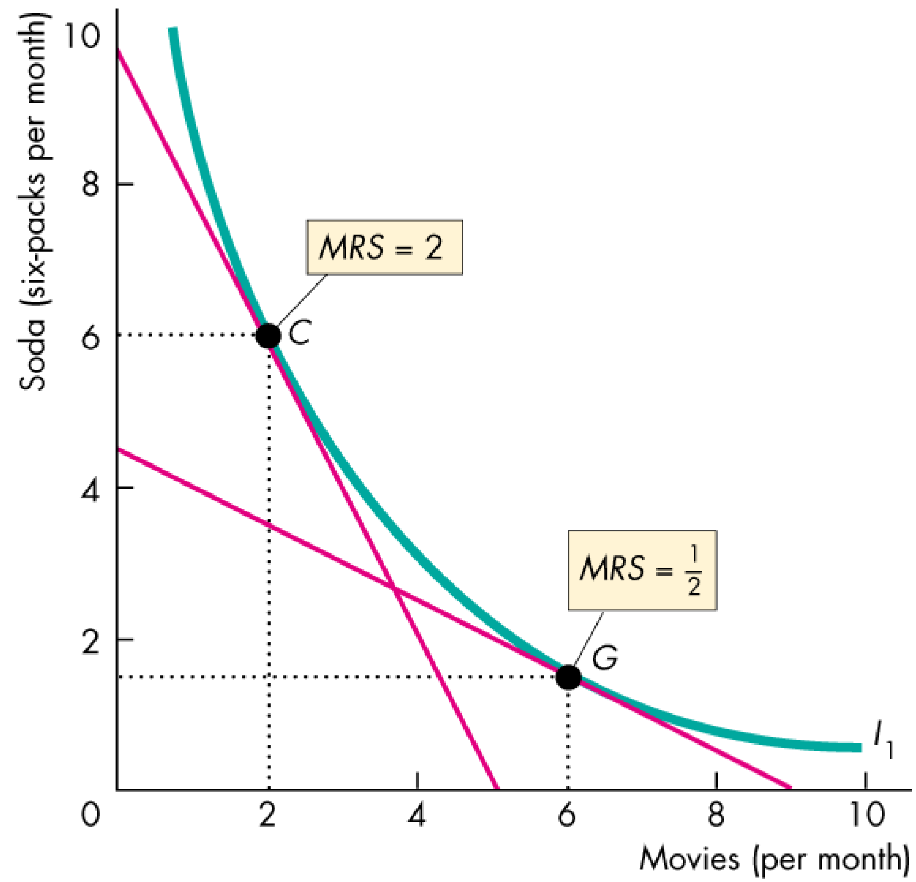
- If the indifference curve is relatively steep, the *MRS* is high. In this case, the person is willing to give up a large quantity of *y* to get a bit more *x*.
- If the indifference curve is relatively flat, the *MRS* is low. In this case, the person is willing to give up a small quantity of *y* to get more *x*.
- A diminishing marginal rate of substitution is the key assumption of consumer theory.
- A **diminishing marginal rate of substitution** is a general tendency for a person to be willing to give up less of good *y* to get one more unit of good *x*, and at the same time remain indifferent, as the quantity of good *x* increases.

Law of diminishing Marginal Rate of Substitution

– Figure shows the diminishing *MRS* of movies for soda.

At point *C*, Lisa is willing to give up 2 six-packs to see one more movie—her *MRS* is 2.

At point *G*, Lisa is willing to give up 1/2 a six-pack to see one more movie—her *MRS* is 1/2.

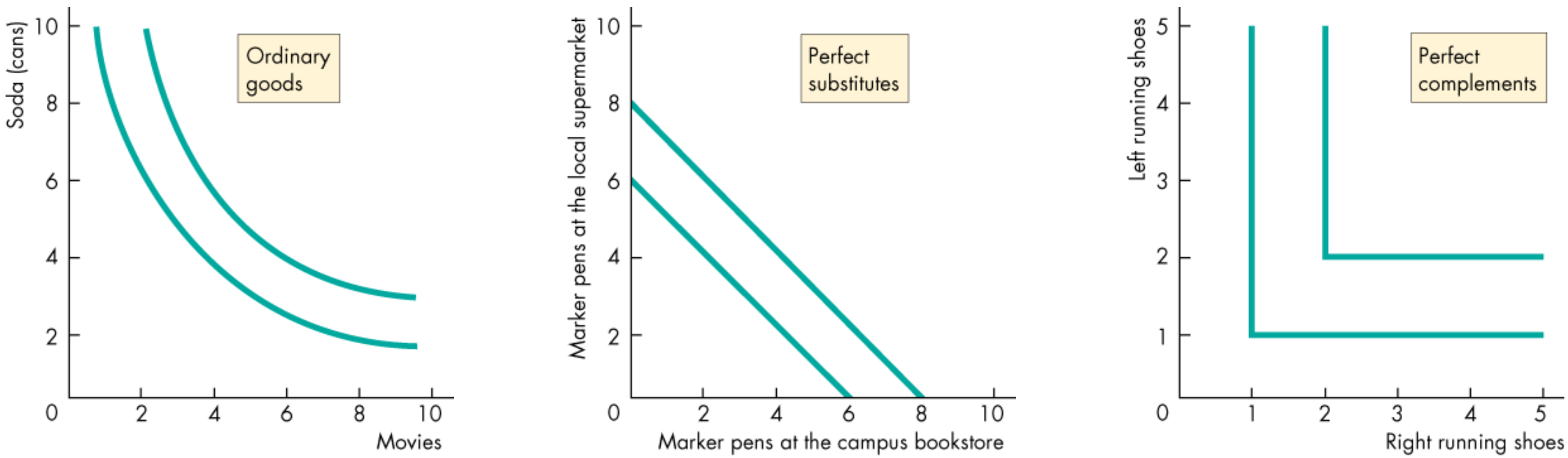


Characteristics of IC:

- It is downward sloping (from left to right)
- It is convex to the origin.
- Higher IC shows a higher level of preference/satisfaction
- ICs do not cross each other.

Different types of Indifference Curve

- Degree of Substitutability
  - The shape of the indifference curves reveals the degree of substitutability between two goods.
  - Figure shows the indifference curves for ordinary goods, perfects substitutes, and perfect complements.



(a) Ordinary goods **(b) Perfect substitutes** **(c) Perfect complements**

Consume Equilibrium

– The consumer's best affordable point is:

- On the budget line
- On the highest attainable indifference curve
- Has a marginal rate of substitution between the two goods equal to the relative price of the two goods

The following condition needs to hold:

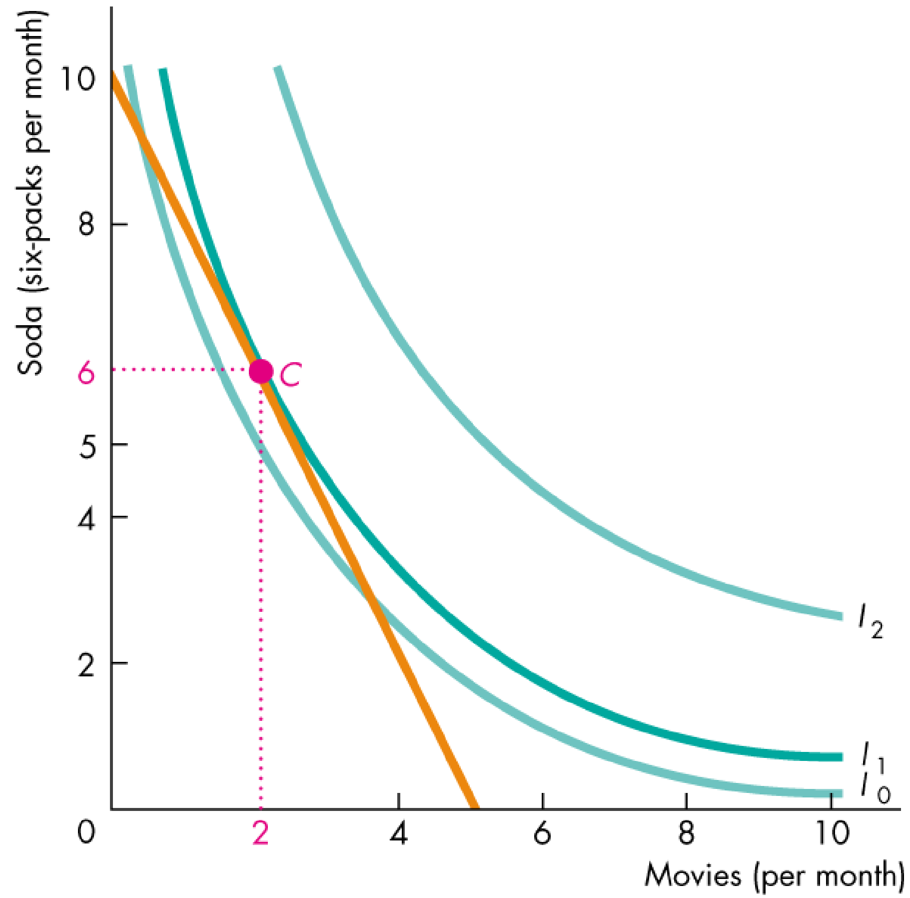
$$\frac{P_X}{P_Y} = \frac{MU_X}{MU_Y} \qquad \frac{MU_X}{P_X} = \frac{MU_Y}{P_Y}$$

– Here, the best affordable point is *C*.

Lisa can afford to consume more soda and fewer movies at point *F*.

And she can afford to consume more movies and less soda at point *H*.

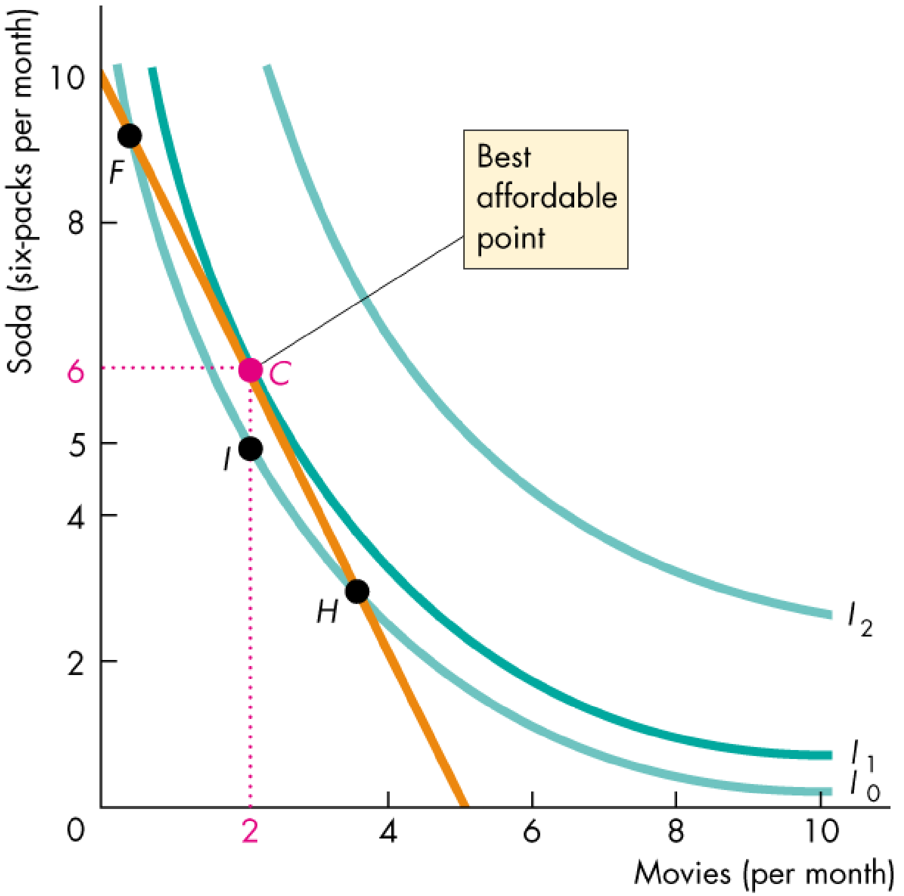
But she is indifferent between *F*, *I*, and *H* and she clearly prefers *C* to *I*.



At point *F*, Lisa's *MRS* is greater than the relative price.

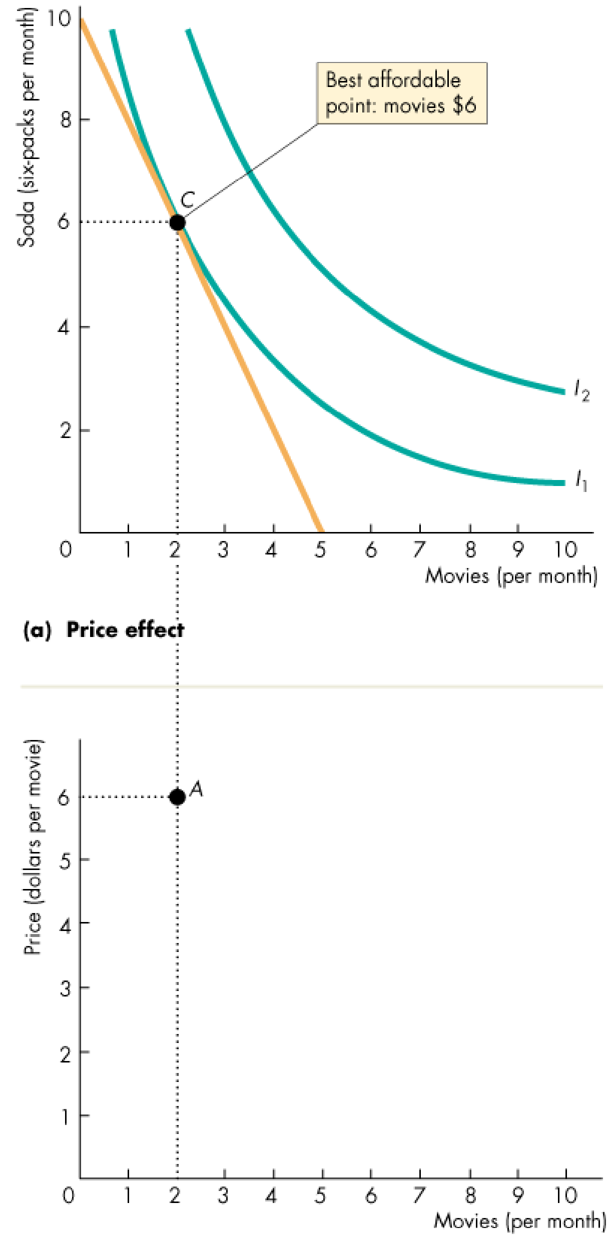
At point *H*, Lisa's *MRS* is less than the relative price.

At point *C*, Lisa's *MRS* is equal to the relative price.



From IC to demand curve

- A Change in Price
  - The effect of a change in the price of a good on the quantity of the good consumed is called the **price effect**.
  - Figure illustrates the price effect and shows how the consumer's demand curve is generated.
  - Initially, the price of a movie is \$6 and Lisa consumes at point *C* in part (a) and at point *A* in part (b).



(a) Price effect

(b) Demand curve

From IC to demand curve

- The price of a movie then falls to $3.

The budget line rotates outward.

Lisa's best affordable point is now *J* in part (a).

In part (b), Lisa moves to point *B*, which is a movement along her demand curve for movies.